Exam 4. Practice 3.

*You may use a basic calculator and formula sheets. Your sheets must contain formulas only, no examples. Phones and all other smart devices are forbidden. Any AI usage will result in an automatic F for the exam. Show **ALL** your work for credits.*

Notice: For all the maximization problems, it is required to do the three-step process to solve the problem.

1. Find $f'(x)$. Simplify 1c.
   a. $f(x) = -2x^3 + 3x + 19$

   b. $f(x) = \frac{2}{x^3} - \frac{4}{x^7} - 5x + 2026$

   c. $f(x) = \frac{3x^3+1}{3x^3-1}$

   d. $f(x) = (2x^2 + 1)(3x^3 + x)$

2.

The per-day cost function (in dollars) for manufacturing a specific high-end electronic component is given by:

$$C(x) = 16,000 + 50x + 2x^2$$

Where x is the number of units manufactured per day ($0 < x \leq 200$).

a. Find the Marginal Cost function.
b. Calculate and interpret the meaning of $C'(40)$.
c. Write the Average Cost function, $AC(x)$.
d. How many items should be manufactured per day in order to minimize the average cost?

3.

A professional pickleball paddle manufacturer has determined that the daily price p (in dollars) they can charge is related to the quantity x of paddles they produce by the following quadratic demand function:

$$p(x) = -x^2 + 60x + 1,500$$

a. Find the revenue function $R(x)$ and the marginal revenue $MR(x)$ function.

b. Interpret the meaning of $MR(10)$.

c. Find the quantity x that should be produced in order to maximize the revenue function.

4.

A company determines that when x units of a product are produced each month, they will be sold at a price of:

$$p = 40 - 2x$$

dollars per unit. The total cost of producing x units is:

$$C(x) = x^2 + 10x + 50$$

a. Find the profit function $P(x)$ and the marginal profit $MP(x)$ function.
b. Interpret the meaning of $MP(4)$.
c. How many units should the company produce to maximize the profit?